Plan Overview

A Data Management Plan created using DMPonline.be

Title: Bioavailability trial of Tomovicen (WPMTX2005)

Creators: Stan Frederickx https://orcid.org/0009-0006-8586-8408, Vincent Renda, Tom Nangosyah https://orcid.org/0009-0003-5995-6621, Selma Bouggar

Affiliation: Université catholique de Louvain (UCLouvain)

Funder: Fonds National de la Recherche Scientifique (FNRS)

Template: UCLouvain

Data Manager: Tom Nangosyah https://orcid.org/0009-0003-5995-6621, Selma Bouggar, Vincent Renda, Stan Frederickx https://orcid.org/0009-0006-8586-8408

Project Administrator: Tom Nangosyah https://orcid.org/0009-0003-5995-6621, Selma Bouggar, Vincent Renda, Stan Frederickx https://orcid.org/0009-0006-8586-8408

Project abstract:

The aim of this project is to evaluate the pharmacokinetic profile of Tomovicen, a drug similar to Tamoxifen, in a Phase I bioavailability study comparing oral and intravenous administration using a 2 × 2 crossover design.

ID: 226575

Start date: 12-10-2026

End date: 26-10-2026

Last modified: 26-10-2026

Lifecycle stage: Intermediate

Collaborators:

Name	ORCID	Role(s)	Organization	ROR
Stan Frederickx	https://orcid.org/0009-0006-8586-8408	Data Manager Project Administrator	Université catholique de Louvain (UCLouvain)	
Vincent Renda		Data Manager Project Administrator	Université catholique de Louvain (UCLouvain)	
Tom Nangosyah	https://orcid.org/0009-0003-5995-6621	Data Manager Project Administrator	Université catholique de Louvain (UCLouvain)	
Selma Bouggar		Data Manager Project Administrator	Université catholique de Louvain (UCLouvain)	

Bioavailability trial of Tomovicen (WPMTX2005)
GDPR - UCLouvain

Collecte et traitement des données à caractère personnel

Collectez-vous ou traitez-vous des données personnelles ?

- Oui

Collectez-vous ou traitez-vous des données personnelles primaires et/ou des données personnelles secondaires ?

- Données personnelles secondaires

We do not collect any primary personal data. We only process secondary personal data originating from Phase I study conducted by our own company. This processing is covered by the participants' initial consent

Si vous traitez des données personnelles secondaires, allez-vous informer les personnes dont les données personnelles sont traitées ou ont-elles déjà été informées ?

- Oui

Dans la négative, expliquez pourquoi il est impossible ou pourquoi cela impliquerait un effort disproportionné d'informer les personnes dont les données personnelles sont traitées.

Not applicable (N/A)

Catégories de données à caractère personnel et personnes concernées

Collectez-vous/traitez-vous l'une des catégories spéciales de données suivantes ?

- Des données qui révèlent l'origine raciale ou ethnique, les opinions politiques, les croyances religieuses ou philosophiques, ou l'appartenance syndicale.
- Des données génétiques

For this project, we received a limited, predefined subset of data from a Phase I study conducted in our own company. This question therefore only covers the data actually transferred to us.

Quelles sont les autres catégories de données personnelles que vous collectez/traitez ?

- Caractéristiques physiques (taille, poids...)
- Caractéristiques personnelles (âge, sexe, date de naissance, statut marital, nationalité...)

For this project, we received a limited, predefined subset of data from a Phase I study conducted in our own company. This question therefore only covers the data actually transferred to us.

À qui appartiennent les données personnelles que vous collectez/traitez ?

- Autres (veuillez préciser ci-dessous)

The data received relate exclusively to adult participant (>18 years). Based on the study design of a Phase I trial and the available documentation, no minors were included. However, we cannot verify from the data itself whether any participants belong to vulnerable individuals.

Votre recherche sera-t-elle sérieusement entravée si les personnes dont les données personnelles sont collectées/traitées exercent leur droit d'accès, de rectification, de limitation du traitement, d'oubli, de portabilité des données et/ou d'opposition ?

- Non

Si oui, veuillez justifier la nécessité de déroger à un ou plusieurs des droits mentionnés à la question 8. Une justification est requise pour chaque dérogation.

Not applicable (N/A)

Finalité(s) du traitement

Quelle(s) est (sont) la (les) finalité(s) du traitement des données personnelles ?

The data is used for scientific research and drug development purposes only, within the clinical development programme of Tomovicen

1. Characterisation of the PK-profile of Tomovicen, after oral and intravenous administration in the Phase I bioavailability trial. This means estimation of primary PK parameters (Cmax, Tmax, AUC, half-life, clearance, volume of distribution of the drug) via non-compartmental analysis.
2. Influence of covariates effects (e.g. age, weight, sex, CYP2D6 genotype, ethnicity, ...) on the PK-parameters to identify sources of interindividual variability, which is important for future dose selection.
3. Compartmental PK-modelling to propose a structural model that can support the design and analysis of subsequent clinical phases.

Quel est le fondement juridique du traitement ?

- Le consentement

Si le traitement est fondé sur une *mission d'intérêt public* ou une *obligation légale*, précisez le motif ou la référence légale.

Not applicable (N/A)

Un comité d'éthique de l'université, ou un comité d'éthique externe, a-t-il pris une décision sur l'étude proposée ? Si oui, veuillez préciser lequel.

Yes for the original trial, but not for this secondary analysis

Responsabilité en matière de RGPD

Y a-t-il une autre université, un autre institut de recherche ou un autre partenaire impliqué dans la recherche (outre l'UCLouvain) ?

- Non : l'UCLouvain est le seul responsable du traitement des données

Not applicable (N/A)

Si oui, veuillez préciser qui détermine les objectifs ("pourquoi") et les moyens ("comment") de la recherche.

Not applicable (N/A)

Cette étude impliquera-t-elle le recours à des sous-traitants pour traiter les données personnelles ?

- Non

Veuillez préciser l'identité de ce sous-traitant

Not applicable (N/A)

Transferts de données et catégories de destinataires

Est-ce que vous divulguez/partagez/transférez des données personnelles au-delà de votre équipe de projet, soit avec des destinataires de l'UCLouvain, soit avec des destinataires externes ?

No, personal data are not disclosed beyond the project team. The dataset received from the Phase I study team is processed exclusively by the members of the analysis project
As an academic exercise, the analysis, report and reproducibility dashboard are submitted to the course instructors of "Integrative Work 1 [WPMTX2005]" at UCLouvain.
Since the data are anonymized, no personal data are shared at all.

Si oui, à ou avec quelles catégories de destinataires les données personnelles sont-elles divulguées/partagées/transférées ?

- Autres (veuillez préciser ci-dessous)

Not applicable (N/A)

Si oui, où les données personnelles sont-elles divulguées/partagées/transférées ?

Not applicable (N/A)

Quelle(s) est (sont) la (les) finalité(s) du transfert de données ?

Not applicable (N/A)

Période de conservation

Quelle est la durée de conservation prévue pour les différentes catégories de données à caractère personnel ? Veuillez expliquer.

Category	Retention period	Rationale
Raw dataset (received from Phase I study team)	Duration of the project	Needed for the verification of the data analysis, study and reproducibility
Derived data and analysis outputs	Duration of the project	Supports reproducibility and the re-use of the reported results and any follow-up modelling work.
Analysis code, documentation and dashboard	Duration of the project	Support the result and making the results more transparent

Mesures de sécurité

Comment les données seront-elles stockées ?

- Autre

si la case "autre" a été cochée, veuillez fournir plus de détails sur ce stockage.

SharePoint (Microsoft 365)

Les données seront-elles rendues anonymes ou pseudo-anonymes ?

- Oui

The data will be anonymized/pseudonymized by the Phase I study team before transfer to our analysis team: all direct identifiers were removed or replaced by anonymous subject codes.

Our team does not hold or received the identification key. No further anonymization step is performed by our team. We preserve the data as received

Si les données n'ont pas été anonymisées ou pseudonymisées (c'est-à-dire si vous travaillez avec des données brutes), veuillez en indiquer brièvement les raisons.

Not applicable (N/A)

Si les données sont conservées sur support papier, indiquez la méthode de stockage.

- Autre

si la case "autre" a été cochée, veuillez fournir plus de détails sur ce stockage.

Not applicable (N/A). No data are stored on paper. The dataset was received as an electronic file (.xlsx) and all analyses, results and documentation are stored and processed exclusively in digital form. We do not make paper printouts containing individual-level data. If any working printout of aggregated results were produced, it would be stored securely and destroyed after use.

Analyse des risques

Pour analyser les risques éventuels liés au traitement des données personnelles, veuillez cocher les cases qui s'appliquent à cette recherche.

- Des catégories particulières de données personnelles sont traitées dans le cadre de cette recherche (voir question 5).

La recherche constitue-t-elle un traitement à haut risque probable ? Si vous avez coché deux cases ou plus à la question 29, la réponse est "oui". Si la réponse est oui, merci de compléter la section "DPIA".

Not applicable (N/A)

Bioavailability trial of Tomovicen (WPMTX2005)
UCLouvain Template

Administrative Data

Date of first version

15/10/2026

Date of last version

26/10/2026

Data collection

What is the purpose of the data collection/generation and its relation to the objectives of the project?

The data are collected in a Phase I 2 × 2 crossover study comparing intravenous and oral administration (500 mg) of a tamoxifen-like prodrug activated by CYP2D6. The aim is to estimate absolute oral bioavailability and characterise the pharmacokinetic profile and its interindividual variability. Additional objectives include exploring covariate effects and developing a compartmental model to inform dose selection and dosing regimens in subsequent stages of development. The data are managed to ensure that all analyses are transparent and reproducible.

To whom might the data be useful ('data utility')?

The data are useful to:

- pharmacometricians and clinical pharmacologists (PK modelling, bioavailability, covariate analysis)
- the sponsor's development team (dose and regimen decisions, CYP2D6 effects)
- researchers on tamoxifen analogues
- regulatory reviewers verifying the PK analysis
- educators or methodologists as a reproducible worked example.

What data will you collect or create?

No new or primary data are collected. Our team processes an existing, pseudonymized dataset received from the Phase I study team of our company (restricted data only needed for the analysis).
Types of data that we received (secondary, pseudonymized):

- Demographic/physiological covariates
  (i.e. subject ID (pseudonymized code), age, sex, weight, albumin/creatinine/CRP concentration, eGFR and CYP2D6 genotype)
- PK-data of oral and IV administration

The exact pseudonymization procedure applied by the Phase I study team before transfer is not documented in the information received by our team. The dataset was transferred to us without any identification key or re-identification information. We therefore treat the data as anonymized for our processing, while recognising that formal guarantees of irreversible anonymization were not provided.
In this context, we made a Data Sharing Agreement that states that the company (sponsor) transfers the anonymized dataset to our team solely for the pharmacometric analysis, with no redistribution and no re-identification.

How will the data be collected or created?

No data will be collected by our team. The dataset was collected prospectively by the Phase I study team of our company during the clinical trial. Our team received the anonymized dataset (datIW1PMX001) as a single file (.xlsx) with different Excel sheets from the study team.
All new data created during the project will be generated through documented scripts (R (version 4.6.1) and MonolixSuite - PKanalix (version 2024R1)). This will be in line with reproducible research principles:

- Import and quality check of the raw dataset, where abnormalities are not corrected manually in the raw data file
- Data formatting via script (especially R), with every transformation recorded in the code. This allows us to generate different traceable versions.
- Derived data
- Outputs: generated from the scripts

We aimed to create a uniform and consistent style throughout the script. We do this using a Quarto file in RStudio with modular pieces of code and sufficient accompanying explanations to ensure the quality.

Data documentation and metadata

How will you document the data?

Documentation of data will be captured at two levels:

1. General data documentation
   A README file needed to understand and reuse the data:
   - Project description: object of the PK-analysis of the Phase I trial of Tomovicen (oral vs intravenous; 2x2 crossover)
   - Origin of data
     - Who?
       The Phase I study team of our company
     - When?
       09-10-2026
     - Why?
       To characterise the PK-profile of Tomovicen, a novel prodrug with structural and mechanistic similarities of Tamoxifen
     - How was data collected?
       All information was recorded per subject, and captured and anonymized by the Phase I study team. This information was transferred to our team as an Excel dataset with multiple sheets without any identification key.
     - Under which condition was the data transferred?
       Anonymized without a key
   - Data quality procedures: anomalies identified and how the data was handled
   - Analysis procedures: the order of the scripts, their purpose and the software versions that is used (R and Monolix)
2. Metadata
   - Data dictionary: for each variable (name, description, type and units of measurement)

Ethical and legal issues

How will you manage any ethics and confidentiality issues?

The original Phase I trial was conducted under the applicable Clinical Trial Regulation including EC approval and ICF from the participants. These ICF covered the use of their data for the clinical development programme of the compound, which includes this secondary pharmacometrical analysis. The data was first formatted into an anonymised form and then transferred to our team without identifiable key. Consequently, the analysis itself raises no further confidentiality risks.

- Raw dataset was generated by Université catholique de Louvain (UC Louvain) during the Phase I trial. The company, as sponsor, owns the intellectual property rights to this dataset. Consequently we will not deposit the data in a public repository or as publication supplements
- Derived data and analysis outputs created by our team, where we hold the authors' rights on the analysis code and outputs

- Analysis code and documentation, where a re-use licence may be applied to the code alone, independently of the data
- In addition no third-party or publicly available data are re-used

How will you manage intellectual property rights issues?

- Raw dataset (datIW1PMX001):
  Generated by Université catholique de Louvain (UC Louvain) during a Phase I bioavailability trial. As sponsor Université catholique de Louvain (UC Louvain) owns all property rights to the dataset. Our team may reuse the data for this pharmacometric analysis (internal reuse with preservation for the analysis) within the drug development programme, with no right to redistribute, publish or deposit the dataset. In addition no licence is applied to the dataset, as it cannot be shared.
- Derived data and outputs:
  Created by our team with corresponding author rights. However, the property rights follow the company's intellectual property policy, as the results derive from the proprietary trial data.

Storage and backup

How will you store and backup data during research?

All project data (raw and adjusted datasets) are stored in SharePoint (Microsoft 365). The dataset is small (low MB), so capacity is sufficient and no additional storage needs to be requested.
Analysis code and documentation are versioned via GitHub.

How will you ensure that stored data are secure?

Risk assessment and measures that we have taken:

1. Accidental modification or loss of the raw dataset
   - Datasets were never sent via personal email or commercial cloud devices
2. Exposure of any residual personal data through insecure transfer
   - Data was received via a secure transfer channel
3. Unauthorised access
   - Access control
   - No access is given to persons outside the project team. If this would be necessary, there is need of approval from the sponsor
   - Data will be deleted from the storage by the group administrator and working copies on personal devices will be removed (everyone is responsible for that)

Data selection and preservation after research

Which data should be retained for preservation and/or sharing?

All project data (raw dataset, curated data, derived results, analysis code and documentation) are retained in full. They will be preserved until the compound obtains market authorisation, as the data may be needed throughout clinical development for re-analysis, modelling updates and regulatory questions. After market authorisation, the responsibility for long-term preservation transfers to the sponsor.

What is the long-term preservation plan for the selected datasets?

- During the retention period:
  The complete reproducibility package is preserved on SharePoint (Microsoft 365). The code is additionally preserved via GitHub, providing long-term versioning and protection against accidental loss
- End of our team's retention period:
  The responsibility transfers to the sponsor at market authorization of the compound

Data sharing

Are any restrictions on data sharing required?

Yes

- The raw dataset contains personal (health-related) data of the trial participants. However it was received anonymized without identification key, the participant's ICF covered use of their data within the clinical development programme of the compound, meaning that this anonymized status alone does not authorize open sharing
- The dataset is owned by the sponsor. Our team's permission only covers internal reuse for this analysis only
- The data concern an investigational compound in active development creating industrial confidentiality duty and patent-related restrictions

In line with the FAIR-principle "as open as possible, as closed as necessary", our team applies the maximum openness within the permitted boundary : fully documented, reproducible and reusable data internally, with public sharing limited to sponsor approval.

How will you share data selected for sharing?

The raw and derived data will not be shared publicly, because it is sponsor property covered by industrial confidentiality and patent-related restrictions
The complete reproducibility package will be internally shared within the sponsor's organisation for the purposes of the drug development programme via internal secure channels.

Responsibilities and resources

Who will be responsible for data management?

Data management responsibilities are distributed within the project team:
Stan Frederickx - DMP owner:
Responsible for implementing the DMP and keeping it up to date
Vincent Renda - Data & storage coordinator:
Responsible for managing the storage and access right, and monitoring backups Selma Bouggar - Documentation and metadata:
Responsible for data dictionary and metadata Tom Nangosyah - Code and reproducibility:
Responsible for version control of the analysis scripts (GitHub), log files and reproducibility dashboard

Will you need additional resources to implement your DMP?

No additional resources are required beyond the existing institutional provision.

- Software: all analyses are performed with R (open-source and freely available) and Monolix (provided by institution)
- Hardware/storage: the dataset is small. Storage, backup and versioning are covered by the standard university resources
- Expertise: team members have the required skills through the courses.
- Data management costs: none